

CISC3024 Pattern Recognition

Homework #1

August, 2026

Solutions of Homework #1 should be written or printed in a hard copy and handed in on class next Monday (Sept. 7, 2026). No Late submission is accepted.

3.1 Consider a case in which class ω_1 consists of two feature vectors $[0, 0]^T$ and $[0, 1]^T$ and class ω_2 of $[1, 0]^T$ and $[1, 1]^T$. Use the perceptron algorithm in its reward and punishment form, with $\rho = 1$ and $w(0) = [0, 0]^T$, to design the line separating two classes.

3.2 Draw three lines in the two-dimensional space

$$x_1 + x_2 = 0$$

$$x_2 = \frac{1}{4}$$

$$x_1 - x_2 = 0$$

For each of the polyhedra that are formed by their intersections, determine the vertices of the cube into which they will be mapped by the first layer of a multilayer perceptron, realizing the preceding lines. Combine the regions into two classes so that

- (a) a two-layer network is sufficient to classify them
- (b) a three-layer network is necessary

For both cases compute analytically the corresponding synaptic weights.